Truth is Conditional

Submission to the Bingham Philosophical Award

ABSTRACT: This paper examines whether recent proposals that introduce content-sensitive epistemic value parameters are required to explain rational disagreement. I argue that once we take seriously the constraints of epistemic grounding and accessibility, the motivation for such parameters weakens. Apparent symmetry cases, where agents sharing the same credences and assessments of informativeness yet diverge in belief, are difficult to sustain under these constraints. I suggest that the phenomena motivating value-based approaches can instead be captured by a more refined account of existing epistemic utility theories, understood as conditional entropy reduction relative to an agent's background epistemic state. On this view, differences that appear to reflect independent epistemic values can be reinterpreted as differences in how propositions interact with the inferential structure encoded in those states. This framework preserves the core insights of Epistemic Utility Theory while providing a more unified account of epistemic significance.

I. Introduction

A central question in epistemology concerns the nature of epistemic value. A prominent view holds that epistemic good is dependent on accuracy alone; how successful an agent thinks depends on how closely their beliefs align with the truth (Joyce 2009; Greaves and Wallace 2006). Simultaneously, Bayesian philosophers approach belief revision using conditionalization,

saying that updating beliefs upon new evidence maximizes epistemic utility under an accuracy-based approach (Greaves and Wallace 2006).

Other popular approaches use a hybrid technique, called Epistemic Utility Theory (EUT). EUT evaluates beliefs in terms of their expected epistemic value, balancing the likelihood of truth with the informativeness of the belief. Commonly, informativeness is understood in terms of the extent to which a proposition eliminates possible answers to a given question. Recently, EUT has been criticized due to important limitations concerning scoring rules and the role of informativeness (Levinstein 2019).

Yan takes it a step further by arguing agents care not only about the truth but also about which truths are more significant or explanatory, motivating the introduction of a content-sensitive utility function. In what follows, I argue that while Yan's diagnosis is correct, her solution is unnecessary: the phenomena she identifies can be captured within the existing framework by refining the notion of informativeness, rather than introducing a new utility term. Purported cases show agents assign equal probability and equal informativeness to competing hypotheses yet arrive at different beliefs. I argue that such cases are less compelling than they appear. In what follows, I detail the landscape of several epistemic theories, resummon the same issues Yan and Levinstein describe in often-accepted examinations, then offer an alternative to Yan's theory, which offers a rephrasing of Dorst's model as opposed to introducing different variables with their own unknown origins.

II. Original EUTs

Some approaches on epistemic value, such as Jim Joyce's probability-focused framework, focused only on accuracy using metrics such as scoring rules. As Levinstein correctly notes, these models fail to understand the context of different questions. Under recent models such as

Dorst and Mandelkern's, guessing and belief formation is a tradeoff between accuracy and informativity— not probability or truth alone (Dorst and Mandelkern 2021). Suppose there is money behind 3 doors, and one can open either one empty door, or two empty doors. Which is more valuable to you? Both actions are equally accurate, but opening two doors is obviously more informative, as it reduces the number of possible doors left to find the goal. The answer is clearer for Dorst. This model is question-sensitive (Dorst and Mandelkern 2021): whether a guess is rational depends on the question being asked (e.g. should I open 1 door or 2?) because informativity is defined relative to the question's partition. This can be quantified as a function on the epistemic value of belief:

$$\mathbb{E}(p) = P(p) \times JQ_p$$

Where $P(p)$ is the truth, Q_p is informativeness (answer elimination), and J is some weight. This is a clear framework which handles questions where informativity differ. But here is the key: when informativeness and probability are equal, how could two epistemic values differ?

III. Utility-Valued Models

Yan writes that there exist such cases. Suppose two doctors find a patient tested positive for a disease. They differ on whether to believe the patient has the disease. Yet it is likely recommended the doctor still treats the disease anyway. Yan argues that distinctions like this are not due to a difference between belief and action, but rather that agents intrinsically hold different values in getting particular truths right (how doctors may choose proactive prescribing versus rounds of testing). She offers to replace informativity JQ_p with a utility value $u(p, w)$, which measures the utility of belief p in a particular world state. Expanding the prior formula gets us:

$$\mathbb{E}(p) = P(p) \times u(p, w)$$

The function $u(p, w) = J^{Q_p} V_p$ represents the epistemic value of believing p in each world w, allowing different kinds of truths to carry different weights based on both informativity and preference. This is valuable in that it provides a structural explanation for differences in belief. $u(p, w)$ can capture both importance of the belief as well as explanatory value it gives, while not overturning the primary advantages of Dorst's model.

While this proposal successfully captures the intuition that not all truths are treated equally, it is not clear that introducing a separate utility function is necessary to account for this phenomenon. The differences Yan identifies can instead be explained in terms of how competing hypotheses constrain the broader structure of uncertainty. That is, what appears as a difference in epistemic value may reflect differences in informativeness once this notion is properly specified.

IV. Discovering the Bigger Problem

At first glance, this appears to suggest that differences in belief arise from preference. Though this requires adding an entirely new dimension, V_p, which Yan agrees should be separated from informativity as a whole: "The problem isn't with particular models of informativity, but with appealing to informativity (so understood) at all. If we take a step back, we can see that it just doesn't seem right to say that B [who believes an audience member's behavior is due to sexism] believes what he does because he cares more about informativity than C [who believes the behavior is due to the presenters' inexperience] (or vice versa)." (Yan 2026, p. 3) This suggests that due to the possibility someone really prefers a belief –which cannot be proven from any probability or the question itself— we are often very uncertain of an agent's epistemic value of p even given all other information.

The central issue is: how is this hidden preference variable calculated, and is it all so different from our informativity and likelihood?

V. A Conditional Account of Informativity

Standard accounts of informativeness based on answer elimination may be too coarse to capture why rational agents may favor different hypotheses. This has been taken to suggest that informativeness alone is insufficient, motivating the introduction of an additional value parameter. But it could also be that we are using the wrong definition of informativity entirely.

Consider a case in which it will be cloudy on Mars. Either answer is not too informative in explaining anything in our lives, as I assume none of us drive UFOs on Mars. This lack of informativeness is not due to a difference in epistemic value, but to the fact that the proposition does not significantly constrain variables relevant to our current model of the world. By contrast, for an 'agent' whose background knowledge includes variables directly affected by Martian weather, the same proposition would reduce uncertainty across a wider range of relevant variables. This difference is not due to arbitrary preference, but to differences in background knowledge and (prior) beliefs which determine which variables are relevant in evaluating a hypothesis. So, while both belief it will rain on Mars and it will not rain are 50% probable (i.e. $\mathrm{P(p) = P(\neg p) = 0.5}$), and the baseline informativeness of the proposition, understood in terms of answer elimination, is minimal ($Q_p \approx 0.0$), the informativeness of the extraterrestrial agent with a prior belief state B may be quite high, $\mathrm{Q(p \mid B) \gg Q(p)}$. The point is not limited to trivial cases of irrelevance; the same structure applies in harder cases where agents disagree about which variables are explanatory, even within a shared domain.

The apparent variability in informativeness is therefore not a matter of inherent valuation differences, but of how a hypothesis interacts with the structure of an agent's background knowledge. As Fantl and McGrath argue in their Train Cases, "you have the same evidence in each case. But in one case you are justified; in the other you are not," (Fantl and McGrath 2002,

p. 1) showing that justification depends on pragmatic context –not just evidence— and this allows pragmatic encroachment to still exist, through encoding the pragmatic circumstances in terms of backgrounds. What appears as a difference in epistemic value is better understood as a difference in how a hypothesis conditions on an agent's existing informational state. In fact, I want to suggest that so-called stable symmetry cases in which two rational agents sharing the same background knowledge assign equal probability/informativeness to competing hypotheses, yet still arrive at different beliefs, are harder to construct than they might initially appear, and perhaps impossible.

VI. No Stable Symmetry Cases

Recent discussions, particularly those following Yan's proposal, suggest that there may be cases in which two agents share identical credences and identical assessments of informativeness, yet nonetheless diverge in their all-things-considered epistemic evaluations. While these cases are often taken to motivate an additional value parameter, I want to suggest that the dialectical pressure here is weaker than it initially appears. My aim is not to offer a decisive refutation of symmetry cases, but rather to show that once we attend more carefully to the constraints on epistemic justification, the space for such cases narrows considerably.

1. Grounding and Epistemic Relevance

Any factor that is supposed to contribute to the epistemic justification of believing p must, at minimum, be grounded in considerations that the agent can (in principle) recognize as epistemically relevant. This is a modest requirement, but it rules out a large class of putative "value differences" that amount to little more than brute preferences. If a purported epistemic factor cannot be traced to the agent's evidential or inferential commitments, it is unclear how it

could play a justificatory role. This does not imply that non-epistemic factors do not exist, only that they do not directly bear on epistemic evaluation.

2. Accessibility and Background States

A related point concerns epistemic accessibility. If a factor genuinely bears on whether an agent ought to believe p, then the agent must have some route –conceptual, inferential, or evidential— by which its relevance can be reflected within the agent's background state. But any such route will inevitably depend on the agent's background epistemic state B: their prior beliefs, inferential dispositions, and the structure of the questions they take to be salient. Thus, even if we allow that agents may differ in the epistemic significance they assign to a proposition, such differences will typically reflect differences in B, broadly construed. This does not deny that external factors may contribute to the formation or reliability of belief, but only that their epistemic significance must be reflected within B in order to bear on rational evaluation.

3. The Apparent Symmetry

Suppose, for the sake of argument, that two agents A_1 and A_2 share the same background state B, and that they assign the same conditional probability and the same conditional informativeness to p. Yan's model requires that they may nonetheless differ in epistemic value. But once we accept the grounding and accessibility constraints, it becomes difficult to see how such divergence could arise. Any epistemically relevant difference in value must be traceable to some feature of B, yet by stipulation B is held fixed. One might object that pragmatic encroachment allows agents with identical evidence to differ in belief due to differences in practical stakes. However, insofar as stakes influence rational belief, they must do so by shaping the agent's B state. Agents who differ in stakes do not, in fact, share the same B, causing the apparent symmetry to dissolve.

4. A More Modest Conclusion

Although the foregoing lines of thinking place significant pressure on the possibility of genuine symmetry cases, the conclusion they support is ultimately a modest one. The point is not that separate epistemic rationales are incoherent or that agents must invariably converge in their doxastic attitudes. Rather, the claim is that once we impose the minimal constraints of grounding and accessibility, the explanatory burden shifts. Any purported difference in epistemic value between agents who share the same background state must either reflect a previously unacknowledged asymmetry in that state or else appeal to a factor that cannot play a justificatory role. In the former case, the symmetry dissolves; in the latter, the factor is epistemically inert.

This leaves little room for the kind of independent value parameter Yan envisions. The phenomena she highlights, that being cases in which agents appear to "care" differently about the same proposition, can be more parsimoniously explained by differences in how their background commitments structure the landscape of belief. Once those commitments are held fixed, there is no remaining epistemic dimension along which the agents can rationally dissent. Thus, while the argument does not rule out all forms of rational disagreement, it does undermine the motivation for positing an additional value term to account for such cases.

VII. Belief as Conditional Structure

If epistemic differences ultimately trace back to differences in background states, then the remaining task is to clarify the structure of those states. In particular, the preceding discussion suggests that informativeness cannot be treated as a primitive notion; its epistemic role depends on how a proposition reorganizes the uncertainty encoded in an agent's prior commitments. This dependence can be made precise by treating informativeness as a reduction in uncertainty.

The most straightforward characterization is that informativeness corresponds to a reduction in uncertainty: the uncertainty before learning p, minus the uncertainty after. This can be formally represented using entropy (Shannon 1948). The choice of entropy is not essential to the argument; any measure of uncertainty satisfying standard coherence constraints would suffice, though Shannon entropy provides a canonical representation. Thus:

$$Q(p|B) = H(Q|B) - H(Q|B,p).$$

Let $Q = \{q_1, \dots, q_n\}$ be the relevant question space, understood as a partition of possible answers (Skyrms 1999). Given a state B, the agent assigns probabilities $P(q_i|B)$, representing how likely each answer is. Uncertainty is highest when probabilities are evenly distributed (e.g., $P = (0.5,0.5)$), and lowest when one answer is overwhelmingly likely (e.g., $P = (0.99,0.01)$). Entropy provides a standard measure of this uncertainty (Shannon 1948):

$$H(Q|B) = -\sum_{i=1}^{n} P(q_i|B) \log P(q_i|B)$$

This represents the agent's uncertainty about the question before learning p. After conditioning on p, the remaining uncertainty is:

$$H(Q|B,p).$$

Thus, the informativeness of p is the amount by which it reduces uncertainty. This formulation preserves Dorst's notion of answer elimination as a special case. When the prior on all answers is uniform, we have:

$$P(q_i|B) = \frac{1}{n}.$$

Substituting into the entropy formula yields:

$$H(Q|B) = -\sum_{i=1}^{n} \frac{1}{n} \log \frac{1}{n}$$

Suppose that conditioning on p eliminates possibilities, leaving k remaining answers, and that the posterior is uniform over those remaining answers. Then:

$$H(Q|B,p) \;=\; log(k).$$

So:

$$Q(p|B) = log(n) - log(k) = \log\left(\frac{n}{k}\right).$$

Thus, informativeness corresponds exactly to answer elimination: the reduction from n possibilities to k. More generally, however, this framework extends answer elimination by allowing non-uniform prior distributions, so that eliminating more probable possibilities yields greater informativeness than eliminating fewer probable ones. The full epistemic model, conditioned on background belief state, is therefore:

$$E(p|B) = P(p|B)\, Q(p|B).$$

VIII. Positive Implications

The strength of this account lies in a fundamental reinterpretation of the phenomenon Yan identifies. Yan is correct that agents appear to privilege certain truths over others, even when probability and informativeness are held fixed. However, this appearance does not reveal a hidden dimension of epistemic value; rather, it reveals a structural feature of belief that has been historically misidentified. What Yan describes as "caring" about a proposition, such as the belief that it rains on Mars, is actually an agent's sensitivity to the inferential weight of that proposition. A truth seems more "important" not because it is inherently more valuable, but because its acceptance would require a massive reorganization of the agent's existing epistemic web, such as a sudden care for another planet's events. Epistemic value is not a separate metric we "add" to a truth; "importance" is simply a measure of how much of the agent's world would have to change to accommodate that truth.

IX. Addressing Objections

In response to the conditional informativeness framework developed so far, I will address several challenges in turn, arguing that the most serious objections either misidentify the framework's commitments or dissolve once the role of background state B is properly understood.

A defender of Yan might protest that the Grounding Requirement ignores the genuine phenomenological "weight" or "feel" of discovering a profound truth, insisting this experience points to an irreducible value term V_p. This objection mistakes a psychological effect for an epistemic property. The present framework does not deny this phenomenology; rather, it provides its underlying structural mechanism. When a newly learned proposition p forces a massive, cascading reorganization of a highly interconnected background state B, it resolves a tremendous amount of latent uncertainty. The intuitive "weight" of a significant truth is simply the psychological experience of a drastic reduction in conditional entropy, $\mathrm{H}(\mathrm{Q}|\mathrm{B}) - \mathrm{H}(\mathrm{Q}|\mathrm{B}, \mathrm{p})$. Yan is correct that some truths feel intrinsically more significant, but she misdiagnoses the source: this feeling is the cognitive reflection of an overturning of a prior distribution. Because this feeling is a byproduct of entropy reduction rather than an independent epistemic variable, it cannot bypass the Grounding Requirement to serve as a justification for belief.

A second objection holds that the framework handles the doctor's treatment case poorly, failing to explain how two doctors facing identical evidence could rationally disagree. The underlying concern is that real-world disagreements persist even when agents appear to share the same data, suggesting that something beyond probability and informativeness must be doing explanatory work. This objection rests on a mischaracterization of the symmetry case. If two agents are stipulated to share the same B state, and therefore assign identical conditional

probabilities $P(p|B)$ and identical conditional informativeness $Q(p|B)$, then their epistemic situation is already fully specified. There is no remaining epistemic parameter left to vary. Any residual disagreement under these conditions is unlikely to be attributed to epistemic factors, since all epistemically relevant structure has been fixed by stipulation, or else the priors are not as equal as you think. As established earlier, non-epistemic factors do not bear on epistemic evaluation in the relevant sense, even if they may influence practical decision-making. It follows that the symmetry case does not demonstrate the need for an additional value parameter V_p; rather, it shows that whatever is driving the remaining divergence lies outside the domain of epistemic justification altogether.

A related objection charges that the framework conflates epistemic and pragmatic reasoning, particularly in cases where a doctor treats despite diagnostic uncertainty. The worry is that such decisions appear to depend on practical stakes —such as the cost of a missed diagnosis— rather than purely epistemic considerations. If so, the framework would fail to distinguish belief from action. This conflation does not occur. The decision to treat is not the importation of external practical values into an epistemic framework; practical considerations have instead been encoded into the way we choose future actions. A physician whose B encodes repeated observations of successful treatment under uncertainty is not acting on preference but on conditioned epistemic judgment. What appears, from the outside, to be pragmatic behavior can instead be understood as the expression of a richly structured epistemic state.

It has further been objected that the conditional informativeness model merely relocates rather than eliminates hidden parameters, replacing Yan's V_p with an equally opaque partition structure for the question space Q. The concern is that the model avoids one unexplained variable only by introducing another. This objection assumes that question space construction floats free

of background beliefs. Under the present framework, it does not. The relevant partition of Q is determined by B: an agent's prior beliefs fix which distinctions are epistemically salient and therefore which partitions are operative. A physician's question space when diagnosing respiratory illness is structured by prior beliefs about symptom clusters, disease prevalence, and treatment outcomes. This is not an arbitrary or hidden parameter but a transparent consequence of the agent's epistemic history concerning the question asked.

Other objections claim that the symmetry argument proves too much, rendering disagreement vacuously explainable, and all belief deterministic. The worry is that any disagreement can simply be attributed to differences in background state, making the account trivial. This misreads the argument's logical structure. The claim that identical background states entail identical rational beliefs is not a tautology but a substantive thesis. It predicts that any genuine rational disagreement will, upon sufficiently detailed examination, reveal asymmetries in prior belief states. This prediction has empirical content: it directs us to identify specific informational differences rather than positing irreducible value divergence. Moreover, the framework provides a positive account of rational persuasion through prior alignment. When agents disagree, the path to resolution lies in identifying and sharing the prior information that drives divergence, thereby bringing background states into closer alignment. In this sense, rational persuasion amounts to a literal interpretation of "seeing someone's side."

On the formulation of $Q(p|B)$, a more technical objection holds that Shannon entropy is insensitive to the structural content of what is eliminated, treating all probability mass equivalently regardless of its epistemic significance. The concern is that two propositions eliminating equal probability mass may differ in explanatory importance, yet entropy would assign them equal informativeness. If correct, this would undermine the claim that conditional

informativeness captures epistemic relevance. This objection can be resolved by attending to the role of the background epistemic state. Because entropy is computed relative to B, the distribution $P(\cdot\,|B)$ already encodes which possibilities are epistemically significant. Eliminating a high-probability region yields greater reduction in uncertainty than eliminating a low-probability one precisely because B assigns those regions different weights. Moreover, the question space Q is not an unstructured list of possibilities, but a partition shaped by prior epistemic commitments—reflecting a network of causal, evidential, and explanatory relations.

The resulting entropy reduction therefore tracks not merely how many possibilities are ruled out, but how extensively the agent's epistemic structure is reorganized. This captures Dorst's notion of answer elimination in the uniform case while extending it to non-uniform settings without introducing a separate value parameter.

Finally, it may be argued that the framework's conclusions are narrower than claimed, applying only within fixed questions and single agents rather than across agents with different background states. The concern is that the model explains internal consistency but not interpersonal disagreement; how can we compare individuals' decisions where their value functions themselves differ? This objection is addressed by the prior alignment account of rational persuasion. The framework does not merely show that each agent is internally consistent relative to their own B; it provides a method for cross-agent evaluation. When agents disagree, their respective $E(p|B)$ values can be compared by tracing the prior divergences that generate them. Rational argument proceeds by surfacing and sharing the information that drives disagreement, bringing background states into closer alignment. In this way, the framework retains normative traction across agents while remaining within a purely epistemic domain.

X. Conclusion

The problem of impartial informativity shows that existing formulations of Epistemic Utility Theory do not fully account for differences in belief among rational agents. Yan proposes that such differences arise from a content-sensitive value term V_p, reflecting intrinsic variation in what agents take to matter epistemically. While this captures an important intuition, it introduces a form of relativity that threatens the unity of epistemic value.

As I have argued, this additional parameter is unnecessary. Apparent differences in epistemic value do not arise from primitive preferences, but from differences in background epistemic states. Once these states are considered, both probability and informativeness become conditional on B, and the variation Yan attributes to V_p is explained as a consequence of how propositions interact with structured prior beliefs. The resulting model,

$$\mathrm{E(p|B) = P(p|B)Q(p|B)},$$

understands epistemic value as the product of truth-likelihood and conditional informativeness, that is, the degree to which a proposition both aligns with the evidence and reduces uncertainty relative to an agent's belief state.

The central insight is that what appears as "preference" is in fact structured belief. Epistemic value is not an additional dimension layered onto belief, but a feature of belief itself when properly understood in its conditional context. Once this structure is made explicit, the need for a separate value parameter disappears, and the explanatory power of Epistemic Utility Theory is preserved while eliminating the need for additional theoretical machinery. More broadly, the framework suggests that expert disagreement is not evidence of irreducible value pluralism, but rather a diagnostic signal pointing toward asymmetries in background states, which in principle

can be identified, compared, and brought into alignment. This does not guarantee convergence in all cases, particularly where prior commitments remain entrenched, but it provides a framework for diagnosing the sources of persistent disagreement.

Works Cited

1. Dorst, Kevin, and Matthew Mandelkern. "Good Guesses." *Philosophy and Phenomenological Research*, vol. 103, no. 3, 2021, pp. 585–607.
2. Fantl, Jeremy, and Matthew McGrath. "Evidence, Pragmatics, and Justification." *The Philosophical Review*, vol. 111, no. 1, 2002, pp. 67–94. *JSTOR*, https://doi.org/10.2307/3182570. Accessed 1 Apr. 2026.
3. Greaves, Hilary, and David Wallace. "Justifying Conditionalization: Conditionalization Maximizes Expected Epistemic Utility." *Mind*, vol. 115, no. 459, 2006, pp. 607–632.
4. Joyce, James M. "Accuracy and Coherence: Prospects for an Alethic Epistemology of Partial Belief." *Philosophical Studies*, vol. 138, no. 1, 2008, pp. 1–27.
5. Levinstein, B.A. "An Objection of Varying Importance to Epistemic Utility Theory." *Philosophical Studies*, vol. 176, 2019, pp. 2919–2931.
6. Shannon, Claude E. "A Mathematical Theory of Communication." *Bell System Technical Journal*, vol. 27, 1948, pp. 379–423.
7. Skyrms, Brian. "Choice and Chance." *Wadsworth*, 2010.
8. Yan, Pacy. "Not All Truths Are Equal." *The Philosophical Quarterly*, 2026, pqag006, https://doi.org/10.1093/pq/pqag006.